



SAFETY DATA SHEET

1. Identification

Important information

*** This Safety Data Sheet is only authorised for use by HP for HP Original products. Any unauthorised use of this Safety Data Sheet is strictly prohibited and may result in legal action being taken by HP. ***

Product identifier

1CQ02Series

Other means of identification

HP A54 Black Printhead

None.

Recommended use

Inkjet printing

Recommended restrictions

None known.

Manufacturer/Importer/Supplier/Distributor information

HP Inc.
1501 Page Mill Road
Palo Alto, CA 94304-1112
United States

Telephone

650-857-1501

HP Inc. health effects line

(Toll-free within the US)

1-800-457-4209

(Direct)

1-760-710-0048

HP Inc. Customer Care Line

(Toll-free within the US)

1-800-474-6836

(Direct)

1-208-323-2551

Email:

sustainability@hp.com

2. Hazard(s) identification

Physical hazards

Not classified.

Health hazards

Sensitization, skin

Category 1

Environmental hazards

Not classified.

OSHA defined hazards

Not classified.

Label elements



Signal word

Warning

Hazard statement

May cause an allergic skin reaction.

Precautionary statement

Prevention

Wear protective gloves/protective clothing/eye protection. Avoid breathing mist/vapor. Contaminated work clothing must not be allowed out of the workplace.

Response

IF ON SKIN: Wash with plenty of water. If skin irritation or rash occurs: Get medical advice/attention. Take off contaminated clothing and wash it before reuse.

Storage

Not available.

Disposal

Dispose of contents/container in accordance with local/regional/national/international regulations.

Hazard(s) not otherwise classified (HNOC)

Carbon black is classified by the IARC as a Group 2B carcinogen (the substance is possibly carcinogenic to humans). Carbon black in this preparation, due to its bound form, does not present this carcinogenic risk. None of the other ingredients in this preparation are classified as carcinogens according to ACGIH, EU, IARC, MAK, NTP or OSHA. Potential routes of overexposure to this product are skin and eye contact. Inhalation of vapor and ingestion are not expected to be significant routes of exposure for this product under normal use conditions. Complete toxicity data are not available for this specific formulation.

Supplemental information

None.

3. Composition/information on ingredients**Mixtures**

Chemical name	Common name and synonyms	CAS number	%
Water		7732-18-5	70-90
Cyclo Amide		Proprietary*	<10
Black Pigment		Proprietary	<5
2,4,7,9-Tetramethyl-5-decyne-4,7-di ol, ethoxylated		9014-85-1	<1
1,2-Benzisothiazolin-3-one (Benzisothiazolinone)		2634-33-5	<0.1
2-Methyl-2h-isothiazol-3-one (Methylisothiazolinone)		2682-20-4	<0.1

Composition comments

This ink supply contains an aqueous ink formulation.
This product has been evaluated using criteria specified in 29 CFR 1910.1200 (Hazard Communication Standard).

*Proprietary

4. First-aid measures**Inhalation**

Move to fresh air. If symptoms persist, get medical attention.

Skin contact

Wash affected areas thoroughly with mild soap and water. If irritation persists get medical attention.

Eye contact

Do not rub eyes. Immediately flush with large amounts of clean, warm water (low pressure) for at least 15 minutes or until particles are removed. If irritation persists get medical attention.

Ingestion

If ingestion of a large amount does occur, seek medical attention.

Most important symptoms/effects, acute and delayed

Not available.

5. Fire-fighting measures**Suitable extinguishing media**

CO2, water, dry chemical, or foam

Unsuitable extinguishing media

None known.

Specific hazards arising from the chemical

Not applicable.

Special protective equipment and precautions for firefighters

Not available.

Specific methods

Wear self contained breathing apparatus for fire fighting if necessary.

6. Accidental release measures**Personal precautions, protective equipment and emergency procedures**

Wear appropriate personal protective equipment.

Methods and materials for containment and cleaning up

Use a non-combustible material like vermiculite, sand or earth to soak up the product and place into a container for later disposal.

Environmental precautions

Do not let product enter drains. Do not flush into surface water or sanitary sewer system.

7. Handling and storage**Precautions for safe handling**

Avoid contact with skin, eyes and clothing.

8. Exposure controls/personal protection

Occupational exposure limits

US. OSHA Table Z-1 Limits for Air Contaminants (29 CFR 1910.1000)

Components	Type	Value
Black Pigment	PEL	3.5 mg/m3

US. OSHA Table Z-3 (29 CFR 1910.1000)

Components	Type	Value	Form
Black Pigment	TWA	5 mg/m3	Respirable fraction.
		15 mg/m3	Total dust.
		50 mppcf	Total dust.
		15 mppcf	Respirable fraction.

US. ACGIH Threshold Limit Values

Components	Type	Value	Form
Black Pigment	TWA	3 mg/m3	Inhalable fraction.

US. NIOSH. Immediately Dangerous to Life or Health (IDLH) Values, as amended

Components	Type	Value
Black Pigment	IDLH	1750 mg/m3

US. NIOSH: Pocket Guide to Chemical Hazards

Components	Type	Value
Black Pigment	TWA	0.1 mg/m3

Biological limit values

No biological exposure limits noted for the ingredient(s).

Exposure guidelines

Exposure limits have not been established for this product.

Appropriate engineering controls

Use in a well ventilated area.

Individual protection measures, such as personal protective equipment

Eye/face protection

Not available.

Skin protection

Hand protection

Not available.

Other

Use personal protective equipment to minimize exposure to skin and eye.

Respiratory protection

Not available.

Thermal hazards

Not available.

General hygiene considerations

Handle in accordance with good industrial hygiene and safety practice.

9. Physical and chemical properties

Appearance

Physical state

Liquid.

Form

Not available.

Color

Black.

Odor

Not available.

Odor threshold

Not available.

pH

8.8 - 9.1

Melting point/freezing point

Not available.

Initial boiling point and boiling range

Not available.

Flash point

>230.0 °F (>110.0 °C) Pensky-Martens Closed Cup

Evaporation rate

Not available.

Flammability (solid, gas)

Not available.

Upper/lower flammability or explosive limits

Explosive limit - lower (%) Not available.

Explosive limit - upper (%) Not available.

Vapor density Not available.

Relative Vapor density Not available.

Solubility(ies)

Solubility (water) Not available.

**Partition coefficient
(n-octanol/water)** Not available.

Auto-ignition temperature Not available.

Decomposition temperature Not available.

Viscosity 3.2 - 3.3 cP

Other information For other VOC regulatory data/information see Section 15.

Kinematic viscosity <3 mm²/s at 25°C (calculated).

VOC 42 g/L US EPA Method 24

9.2. Other information

Density 1.04 g/cm³

10. Stability and reactivity

Reactivity Not available.

Chemical stability Stable under recommended storage conditions.

Possibility of hazardous reactions Will not occur.

Conditions to avoid Not available.

Incompatible materials Incompatible with strong bases and oxidizing agents.

Hazardous decomposition products Upon decomposition, this product may yield gaseous nitrogen oxides, carbon monoxide, carbon dioxide and/or low molecular weight hydrocarbons. Decomposition of this product may yield oxides of phosphorus.

11. Toxicological information**Information on likely routes of exposure**

Inhalation Under normal conditions of intended use, this material is not expected to be an inhalation hazard.

Skin contact May cause sensitization by skin contact.

Eye contact Contact with eyes may result in mild irritation.

Ingestion Health injuries are not known or expected under normal use.

Symptoms related to the physical, chemical and toxicological characteristics Not available.

Information on toxicological effects

Acute toxicity Based on available data, the classification criteria are not met.

Components	Species	Test Results
1,2-Benzisothiazolin-3-one (Benzisothiazolinone) (CAS 2634-33-5)		
Acute		
Dermal		
LD50	Rat	> 2000 mg/kg (OECD 402)
Oral		
LD50	Mouse	1150 mg/kg
	Rat	1020 mg/kg
		670 mg/kg (OECD 401)

Components	Species	Test Results
2,4,7,9-Tetramethyl-5-decyne-4,7-diol, ethoxylated (CAS 9014-85-1)		
Acute		
Inhalation		
LC50	Rat	20 mg/l, 1 h Dusts, mists and fumes. The data are derived from the evaluations or test results achieved with similar products (conclusion by analogy).
Oral		
LD50	Rat	6300 mg/kg
2-Methyl-2h-isothiazol-3-one (Methylisothiazolinone) (CAS 2682-20-4)		
Acute		
Dermal		
LD50	Rat	242 mg/kg (OECD 402)
Inhalation		
LC50	Rat	0.11 mg/l, 4 h (OECD 403)
Oral		
LD50	Rat	120 mg/kg
Black Pigment		
Acute		
Oral		
LD50	Rat	> 10000 mg/kg
Skin corrosion/irritation	Based on available data, the classification criteria are not met.	
Irritation Corrosion - Skin		
2-Methyl-2h-isothiazol-3-one (Methylisothiazolinone)	Corrosive, rabbit (OECD 404)	
1,2-Benzisothiazolin-3-one (Benzisothiazolinone)	Irritating (4 h, rabbit)	
2,4,7,9-Tetramethyl-5-decyne-4,7-diol, ethoxylated	Not irritating, The data are derived from the evaluations or test results achieved with similar products (conclusion by analogy). Test Duration: 24 h	
Serious eye damage/eye irritation	Based on available data, the classification criteria are not met.	
Eye		
1,2-Benzisothiazolin-3-one (Benzisothiazolinone)	Causes serious eye damage (rabbit)	
2-Methyl-2h-isothiazol-3-one (Methylisothiazolinone)	Corrosive, based on OECD 404 results	
2,4,7,9-Tetramethyl-5-decyne-4,7-diol, ethoxylated	Risk of serious damage to eyes, The data are derived from the evaluations or test results achieved with similar products (conclusion by analogy). Species: Rabbit	
Respiratory or skin sensitization		
Respiratory sensitization	Based on available data, the classification criteria are not met.	
Skin sensitization	May cause an allergic skin reaction.	
Sensitization		
2,4,7,9-Tetramethyl-5-decyne-4,7-diol, ethoxylated	May cause sensitization by skin contact.	
Skin sensitization		
1,2-Benzisothiazolin-3-one (Benzisothiazolinone)	Causes sensitization (Guinea pig, OECD 406)	
2-Methyl-2h-isothiazol-3-one (Methylisothiazolinone)	Sensitizing, mice (OECD 429), Sensitizing, guinea pigs (OECD 406)	
Germ cell mutagenicity	Based on available data, the classification criteria are not met.	
Carcinogenicity	Based on available data, the classification criteria are not met.	
Carbon black is classified as a carcinogen by the IARC (possibly carcinogenic to humans, Group 2B) and by the State of California under Proposition 65. In their evaluations of carbon black, both organizations indicate that exposure to carbon black, per se, does not occur when it remains bound within a product matrix, specifically, rubber, ink, or paint. Carbon black is present only in a bound form in this preparation.		

Carcinogenicity

2,4,7,9-Tetramethyl-5-decyne-4,7-diol, ethoxylated

Ames (OECD 471), Chromosomal aberration (OECD 473), gene mutation (OECD 476), negative. The data are derived from the evaluations or test results achieved with similar products (conclusion by analogy).

IARC Monographs. Overall Evaluation of Carcinogenicity

Not listed.

OSHA Specifically Regulated Substances (29 CFR 1910.1001-1053)

Not listed.

US. National Toxicology Program (NTP) Report on Carcinogens

Black Pigment (CAS Proprietary)

Known To Be Human Carcinogen.

Reproductive toxicity

Based on available data, the classification criteria are not met.

Specific target organ toxicity - single exposure

Based on available data, the classification criteria are not met.

Specific target organ toxicity - repeated exposure

Based on available data, the classification criteria are not met.

Aspiration hazard

Based on available data, the classification criteria are not met.

Further information

Complete toxicity data are not available for this specific formulation

12. Ecological information**Ecotoxicity**

Product		Species	Test Results
1CQ02Series			
Aquatic			
<i>Acute</i>			
Fish	LC50	Fathead minnow (<i>Pimephales promelas</i>)	> 750 mg/l, 96 h
Components		Species	Test Results
1,2-Benzisothiazolin-3-one (Benzisothiazolinone) (CAS 2634-33-5)			
<i>Acute</i>			
	EC50	Activated sludge	12.8 mg/l, 3 h (OECD 209)
Other	EC50	<i>Pseudokirchnerella subcapitata</i>	0.11 mg/l, 72 h OECD (201)
	NOEC	<i>Pseudokirchnerella subcapitata</i>	0.055 mg/l, 72 h (OECD 201)
Aquatic			
<i>Acute</i>			
Crustacea	EC50	<i>Daphnia magna</i>	4.4 mg/l, 48 h 2.9 mg/l, 48 h (OECD 202)
Fish	LC50	<i>Oncorhynchus mykiss</i>	2.15 mg/l, 96 h (OECD 203) 0.8 mg/l, 96 h
2,4,7,9-Tetramethyl-5-decyne-4,7-diol, ethoxylated (CAS 9014-85-1)			
<i>Acute</i>			
	LC50	<i>Scophthalmus maximus</i> (turbot)	52 mg/l, 96 h
Aquatic			
<i>Acute</i>			
Crustacea	EC50	<i>Daphnia magna</i>	88 mg/l, 48 h The data are derived from the evaluations or test results achieved with similar products (conclusion by analogy). (OECD 202).
	LC50	<i>Acartia tonsa</i>	166 mg/l, 48 h
Fish	LC50	<i>Pimephales promelas</i>	36 mg/l, 96 h The data are derived from the evaluations or test results achieved with similar products (conclusion by analogy). (OECD 203).

Components	Species		Test Results
2-Methyl-2h-isothiazol-3-one (Methylisothiazolinone) (CAS 2682-20-4)			
Acute			
	EC50	Activated sludge	34.6 mg/l (DIN 38412-3)
Other	EC50	Pseudokirchnerella subcapitata	0.445 mg/l, 120 h (OECD 201)
Aquatic			
Acute			
Crustacea	EC50	Daphnia magna	1.68 mg/l, 48 h (OECD 202)
Fish	LC50	Rainbow Trout	6 mg/l, 96 h (OECD 203)
Chronic			
Crustacea	NOEC	Daphnia magna	0.0442 mg/l, 21 d (OECD 211)
Fish	NOEC	Oncorhynchus mykiss	4.93 mg/l, 98 d (OECD 210)
Persistence and degradability	Not available.		
Biodegradability			
Percent degradation (Aerobic biodegradation-ready)			
1,2-Benzisothiazolin-3-one (Benzisothiazolinone)	85 %, Not readily biodegradable (OECD 301C) Test Duration: 63 d		
2-Methyl-2h-isothiazol-3-one (Methylisothiazolinone)	54.1 %, (OECD 301B) Test Duration: 29 d		
Bioaccumulative potential	Not available.		
Partition coefficient n-octanol / water (log Kow)			
1,2-Benzisothiazolin-3-one (Benzisothiazolinone)	0.7 (OECD 117)		
2-Methyl-2h-isothiazol-3-one (Methylisothiazolinone)	-0.32 (OECD 107)		
Bioconcentration factor (BCF)			
1,2-Benzisothiazolin-3-one (Benzisothiazolinone)	6.62, (OECD 305) Species: Bluegill (Lepomis macrochirus)		
2-Methyl-2h-isothiazol-3-one (Methylisothiazolinone)	48.1, Viscera (1972) Species: Bluegill (Lepomis macrochirus) 5.75, Carcass (1972) Species: Bluegill (Lepomis macrochirus)		
Mobility in soil	Ethoxylated 2,4,7,9 -tetramethyl-5-decyn-4,7-diol: log Koc 1.251 (20 °C, KOCWIN v2.0, MCI)		
Adsorption			
Soil/sediment sorption - log Koc			
1,2-Benzisothiazolin-3-one (Benzisothiazolinone)	0.97, (OECD 121)		
Other adverse effects	Not available.		

13. Disposal considerations

Disposal instructions	Do not allow this material to drain into sewers/water supplies. Dispose of waste material according to Local, State, Federal, and Provincial Environmental Regulations. HP's Planet Partners (trademark) supplies recycling program enables simple, convenient recycling of HP original inkjet and LaserJet supplies. For more information and to determine if this service is available in your location, please visit http://www.hp.com/recycle .
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14. Transport information

DOT	
UN number	Not available.
UN proper shipping name	Not Regulated
Transport hazard class(es)	
Class	Not available.
Subsidiary risk	-
Packing group	Not available.
Environmental hazards	
Marine pollutant	No
Special precautions for user	Not available.
DOT Supplemental Information	DOT Classification only applies to shipments within the US and Puerto Rico.

IATA

UN number	Not available.
UN proper shipping name	Not Regulated
Transport hazard class(es)	
Class	Not available.
Subsidiary risk	-
Packing group	Not available.
Environmental hazards	No
Special precautions for user	Not available.

IMDG

UN number	Not available.
UN proper shipping name	Not Regulated
Transport hazard class(es)	
Class	Not available.
Subsidiary risk	-
Packing group	Not available.
Transport hazard class(es)	
Marine pollutant	No
EmS	Not available.
Special precautions for user	Not available.

ADR

UN number	Not available.
UN proper shipping name	Not Regulated
Transport hazard class(es)	
Class	Not available.
Subsidiary risk	-
Hazard No. (ADR)	Not available.
Tunnel restriction code	Not available.
Packing group	Not available.
Environmental hazards	No
Special precautions for user	Not available.

Further information Not a dangerous good under DOT, IATA, ADR, IMDG, or RID.

Transport in bulk according to Annex II of MARPOL73/78 and the IBC code: Not applicable.

15. Regulatory information

US federal regulations US TSCA 12(b): Does not contain listed chemicals.

Toxic Substances Control Act (TSCA)**TSCA Section 12(b) Export Notification (40 CFR 707, Subpt. D)**

2-Methyl-2h-isothiazol-3-one (Methylisothiazolinone) 1.0 % One-Time Export Notification only.
(CAS 2682-20-4)

CERCLA Hazardous Substance List (40 CFR 302.4)

Not listed.

SARA 304 Emergency release notification

Not regulated.

OSHA Specifically Regulated Substances (29 CFR 1910.1001-1053)

Not listed.

Superfund Amendments and Reauthorization Act of 1986 (SARA)**SARA 302 Extremely hazardous substance**

Not listed.

SARA 311/312 Hazardous chemical Yes

Classified hazard categories Respiratory or skin sensitization

Other federal regulations

Clean Air Act (CAA) Section 112 Hazardous Air Pollutants (HAPs) List

No intentionally added HAP substances.

Clean Air Act (CAA) Section 112(r) Accidental Release Prevention (40 CFR 68.130)

Not regulated.

Safe Drinking Water Act (SDWA)

Not regulated.

Other information

VOC content (less water, less exempt compounds) = 276 g/L (U.S. requirement, not for emissions)
VOC data based on formulation (Organic compounds minus solids)
US EPA Method 24

Regulatory information

HP complies with chemical regulatory requirements in chemical substance notification laws, where applicable. All chemical substances are notified or exempt from notification or listed in the inventory as existing substances in the following countries: US (TSCA), Canada (DSL/NDSL), Australia (AICC), Japan (ISHL, ENCS), Philippines (PICCS), New Zealand (NZIoC), Russia and China (IECSC). For guidance on importation and/or additional requirements for registration schemes such as EAEU, EU, South Korea, Turkey, UK, India and Taiwan, please contact the Sustainability and Compliance Center (sustainability@hp.com).

16. Other information, including date of preparation or last revision

Issue date

29-Apr-2019

Revision date

17-Oct-2025

Version

12

Other information

This SDS was prepared in accordance with USA OSHA Hazard Communications regulation (29 CFR 1910.1200).

Disclaimer

This Safety Data Sheet document is provided without charge to customers of HP. Data is the most current known to HP at the time of preparation of this document and is believed to be accurate. It should not be construed as guaranteeing specific properties of the products as described or suitability for a particular application. This document was prepared to the requirements of the jurisdiction specified in Section 1 above and may not meet regulatory requirements in other countries.

This safety data sheet is meant to convey information about HP inks (toners) provided in HP Original ink (toner) supplies. If our Safety Data Sheet has been provided to you with a refilled, remanufactured, compatible or other non-HP Original supply please be aware that the information contained herein was not meant to convey information about such products and there could be considerable differences from information in this document and the safety information for the product you purchased. Please contact the seller of the refilled, remanufactured or compatible supplies for applicable information, including information on personal protective equipment, exposure risks and safe handling guidance. HP does not accept refilled, remanufactured or compatible supplies in our recycling programs.

Revision information

1. Product and Company Identification: Product Registration Numbers
Regulatory information: Regulatory information

Explanation of abbreviations

ACGIH	American Conference of Governmental Industrial Hygienists
Acute Tox.	Acute toxicity
Aquatic Acute	Short-term (acute) aquatic hazard
Aquatic Chronic	Long-term (chronic) aquatic hazard
Asp. Tox.	Aspiration hazard
Carc.	Carcinogenicity
CAS	Chemical Abstracts Service
CERCLA	Comprehensive Environmental Response Compensation and Liability Act
CFR	Code of Federal Regulations
COC	Cleveland Open Cup
DOT	Department of Transportation
EPCRA	Emergency Planning and Community Right-to-Know Act (aka SARA)
Eye Dam.	Serious eye damage
Eye Irrit.	Eye Irritation
Flam. Liq.	Flammable liquids
Flam. Sol.	Flammable solids
Lact.	Effects on or via lactation
Muta.	Germ cell mutagenicity
IARC	International Agency for Research on Cancer
NIOSH	National Institute for Occupational Safety and Health
NTP	National Toxicology Program
OSHA	Occupational Safety and Health Administration
Ox. Liq.	Oxidising liquids
Ozone	Hazardous to the ozone layer
PEL	Permissible Exposure Limit
Press. Gas	Gases under pressure
RCRA	Resource Conservation and Recovery Act
REC	Recommended
REL	Recommended Exposure Limit
Repr.	Reproductive toxicity
Resp. Sens.	Respiratory sensitization
SARA	Superfund Amendments and Reauthorization Act of 1986
Skin Corr.	Skin corrosion
Skin Irrit.	Skin irritation
Skin Sens.	Skin sensitization
STEL	Short-Term Exposure Limit
STOT RE	Specific target organ toxicity - repeated exposure
STOT SE	Specific target organ toxicity - single exposure
TCLP	Toxicity Characteristics Leaching Procedure
TLV	Threshold Limit Value
TSCA	Toxic Substances Control Act